1. (40 points) Using the truth table to show that the following propositions are tautologies:

a) $p \to (q \to r) \Leftrightarrow p \wedge q \to r$

b) $(p \to q) \wedge (\neg p \to q) \to q$.

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2. (40 points) i) Prove the following identity: $A - (B \cup C) = (A - B) \cap (A \cup C)$

ii) In a survey of university students, 64 had taken mathematics course, 94 had taken chemistry course, 58 had taken physics course, 28 had taken mathematics and physics, 26 had taken mathematics and chemistry, 22 had taken chemistry and physics course, and 14 had taken all the three courses. Draw Venn diagram to represent these data. Find how many had taken physics only.

3. (40 points) i) For each of the following predicates, find all the values in the domain $D = \{1,2,3,4,5,6,7,8,9\}$ such that the predicate is true:

   a) $Q(n) : (n^2 < 50) \wedge (n \text{ is an even number})$

   b) $S(n) : (2n > 7) \vee (n \text{ is divisible by } 3)$

   ii) Given that the domain of discourse is $\mathbb{R} \times \mathbb{R}$. Determine the truth of the statement $\forall y \exists x (x^2 < y + 1)$. (Explain your answer in detail)

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4. (40 points) Which of the following functions are injective? surjective? Explain.
   - $f : \mathbb{N} \to \mathbb{N}, f(x) = \begin{cases} x-1, x \equiv 1 \bmod 2 \\ x+1, x \equiv 0 \bmod 2 \end{cases}$ where $\mathbb{N} = \{0,1,2,3,...\}$
   - $g : \mathbb{Z} \to \mathbb{Z}, g(x) = 3x+1$
   - $h : \{0,1,2,3,4\} \to \{0,1,2,3,4\}, h(x) = 2x+1 \bmod 5$

5. (30 points) Find the least integer n such that f(x) is $O(x^n)$ for each of the following functions:

a) $f(x) = 3x^3 + 4x^2 \log x$

b) $f(x) = (x + 2\log(x+1))^3 + x^3 \log x$

6. (30 points) Prove that for every real numbers x, y the following inequalities holds:

$$\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$$

7. (40 points) i) Show the steps of using Euclidean algorithm to find gcd(9775,5580)
   ii) Find the octal representation of your Troy Id number. (showing detail computation)

8. (40 points) Show the detail steps of using modular exponentiation algorithm to find $106^{109} \bmod 171$